

SAMBP Technical Report TR-82/2028

Cyber-Resilient GOOSE: Anomaly Detection and IEC 62351-6 Security
Isolation Forest Detector, HMAC-SHA256 Authentication, Four Attack Types, `local_io_mode`
Fallback: 10 Tests PASS

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1 Background and Objectives

1.1 Motivation

GOOSE (Generic Object Oriented Substation Event) messaging is the real-time communication backbone of IEC 61850 substations, used for protection trip commands, interlocking, and status exchange. Because GOOSE operates at Layer-2 Ethernet without TCP/IP, it is fast (≤ 4 ms) but also vulnerable to cyber attacks: replay attacks, timing floods, sequence manipulation, and content injection can cause spurious trips or denial-of-service [1].

The SAMBP Digital Twin adds a cyber-resilient GOOSE layer based on: (1) a machine-learning anomaly detector (Isolation Forest) for traffic analysis, and (2) IEC 62351-6:2020 HMAC-SHA256 message authentication.

1.2 Objectives

1. Implement a 5-feature GOOSE anomaly detector with Isolation Forest.
2. Detect all four attack types: `REPLAY`, `TIMING_FLOOD`, `SEQUENCE_SKIP`, `CONTENT_ANOMALY`.
3. Achieve false-positive rate $< 1\%$ on normal traffic.
4. Implement IEC 62351-6 HMAC-SHA256 authentication with key management.
5. Validate `local_io_mode` fallback after key failures.
6. Pass 10 unit tests.

2 Theory and Methodology

2.1 GOOSE Anomaly Detector

2.1.1 Feature Vector

The anomaly detector uses a 5-dimensional feature vector extracted from each GOOSE message:

Table 1: GOOSE anomaly detector: 5-feature vector.

Feature	Description	Attack sensitivity
f_0	$ \Delta \text{stnum} $ (state number change)	<code>REPLAY</code> , <code>SEQUENCE_SKIP</code>
f_1	$ \Delta \text{sqnum} - 1 $ (sqnum deviation from +1)	<code>SEQUENCE_SKIP</code>
f_2	$\log(\text{inter-arrival}/\text{retx_period} + 1)$	<code>TIMING_FLOOD</code>
f_3	Shannon entropy of dataset bytes	<code>CONTENT_ANOMALY</code>
f_4	$\log(1 + \text{numeric range of dataset})$	<code>CONTENT_ANOMALY</code>

2.1.2 Attack Taxonomy and Rules

Rule-based detection handles structured attacks before the ML layer:

REPLAY: Message age > 5 s OR stnum regression ($\text{stnum}_k < \text{stnum}_{k-1}$).

TIMING_FLOOD: Inter-arrival time < 0.5 ms (below minimum GOOSE retransmission interval).

SEQUENCE_SKIP: Discontinuity in stnum or sqnum beyond expected transitions.

CONTENT_ANOMALY: Dataset numeric range $> 3\sigma$ above the training bound OR Isolation Forest outlier score.

2.1.3 Isolation Forest

The Isolation Forest is trained on $n = 263$ normal GOOSE messages (contamination parameter = 0.005, i.e., 0.5% expected outlier fraction). The trained forest assigns an anomaly score $s \in [0, 1]$ to each message; $s > 0.6$ is flagged as anomalous.

2.2 IEC 62351-6 HMAC-SHA256 Authentication

The HMAC-SHA256 message authentication code is computed over:

$$\text{MAC} = \text{HMAC-SHA256}(\text{gocb_ref} \parallel \text{app_id}_{(4B)} \parallel \text{stnum}_{(4B)} \parallel \text{sqnum}_{(4B)} \parallel t_{\text{msg}(8B)} \parallel \text{dataset_bytes}) \quad (1)$$

Key management implements:

- **Active key:** Current signing/verification key.
- **Pending key:** Next key during grace period (both accepted).
- **Grace period:** 60s overlap during key rotation.
- **Fallback:** `local_io_mode` asserted after `fail_threshold = 4` consecutive wrong-key failures. In `local_io_mode`, the IED ignores remote GOOSE commands and operates from hardwired I/O only.

3 Simulation Results

3.1 Anomaly Detector: Normal Traffic

False-positive rate on 263 normal GOOSE messages: 1 false positive $\rightarrow$ FP rate = 0.38% ($< 1\%$ threshold). ✓

3.2 Attack Detection Results

Table 2: Attack detection results.

Attack type	Detected	Severity	Detection method
REPLAY	✓Yes	CRITICAL	Rule: stnum regression
TIMING_FLOOD	✓Yes	CRITICAL	Rule: inter-arrival < 0.5 ms
SEQUENCE_SKIP	✓Yes	HIGH	Rule: sqnum gap
CONTENT_ANOMALY	✓Yes	HIGH	Isolation Forest

3.3 HMAC-SHA256 Authentication

Table 3: HMAC authentication scenarios.

Scenario	HMAC valid	Alarm	Action
Sign + verify (correct key)	True	OK	None
Tampered dataset	False	WARN	Discard
Unsigned message	False	WARN	Discard
4 wrong-key failures	—	CRITICAL	<code>local_io_mode=True</code>

3.4 10-Test Pass Summary

T01–T04: Rule-based detection (one test per attack type). ✓

T05: Isolation Forest false-positive rate $< 1\%$. ✓

T06: HMAC sign/verify (correct key). ✓

T07: HMAC tampered dataset detection. ✓

T08: Unsigned message detection. ✓

T09: Key rotation grace period. ✓

T10: `local_io_mode` after 4 wrong-key failures. ✓

Total: 10/10 PASS.

Finding 1 (Hybrid rule + ML detection is effective). *Rule-based detection handles structured attacks (REPLAY, TIMING_FLOOD, SEQUENCE_SKIP) with zero latency, while the Isolation*

Forest handles novel CONTENT_ANOMALY attacks not covered by explicit rules. False-positive rate 0.38% (263 normal messages, 1 false positive) is well within the $< 1\%$ threshold.

Finding 2 (HMAC-SHA256 authentication prevents content injection). *All tampered GOOSE messages (modified dataset bytes) are correctly detected by HMAC verification failure. The `local_io_mode` fallback after 4 wrong-key failures ensures that a key compromise does not create a denial-of-service condition.*

4 Conclusion

TR-82 delivers the cyber-resilient GOOSE module for the SAMBP Digital Twin. The hybrid rule + Isolation Forest anomaly detector achieves 0.38% false-positive rate and detects all four attack types. IEC 62351-6 HMAC-SHA256 authentication with key rotation and `local_io_mode` fallback is validated in 10 unit tests.

References

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